

INSTRUCTIONS TO CANDIDATES

- 1) Candidate should ensure that all answer books including supplementary answers books received by them bear the signature of the junior supervisor, otherwise the answer books will not be examined.
- 2) Begin the answer to each question on a new page. For each answer, write the corresponding question number and sub question number if any in the respective column provided in the answer sheet.
- 3) Do not write anything in the column provided for the marks to be assigned by the examiners.
- 4) Candidate will not be permitted to leave the examination hall until an hour after the question papers are distributed.
- 5) Every candidate present must sign against their seat number on the attendance sheet provided by the junior supervisor.
- 6) Candidates are forbidden to (i) bring any book, notes. Scribbling papers, mobile telephones, programmable calculators, or any other similar devices.(ii) speak or communicate in any manner to any other candidate, while the examination is in progress, and (iii) take with them any answer-book written or blank while leaving the examination hall. The supervisors/authorized persons are authorized to check the students.
- 7) Candidate suspected to be guilty of any of the aforesaid acts will be allowed to write their paper only after giving an undertaking in writing that the decision of the College Examination Authorities in respect of the reported act of unfair means is binding on them.
- 8) Candidates should write their answers legibly. They are warned that **zero marks** will be assigned to **answers which cannot be assessed by the examiners owing to illegible handwriting**.
- 9) Write on both side of a page. Rough work, when necessary, should be done on the Left- hand side and in pencil only.
- 10) While underlining of answers for focusing attention is permitted, use of varied inks, except for illustration and figures must be avoided.
- 11) No sheet shall be torn from the answer-books provided nor shall additional papers attached to them.
- 12) The answer-books will be scrutinized before they are sent to examiners.
- 13) All answer-books supplied shall be returned whether written or blank.
- 14) Nothing shall be written on the question-paper.
- 15) If candidate needs anything, they should approach the Junior Supervisor without disturbing other candidates. However, they should not leave their seat on my account.
- 16) A warning bell will be given ten minutes before the close of the examination. Candidates will not be allowed to leave the examination hall during the last ten minutes. At the final bell, they must stop writing and be ready to hand over their answer-books to the Junior Supervisor. They should not leave their seats until answers-books from all candidates are collected by the Junior Supervisor.
- 17) A candidate who disobeys any instructions issued by the Senior/Junior Supervisor or who is guilty of rude or disobedient behavior is liable for disciplinary action to be taken against him/her by the College Examination Authorities.

Space
For
MarksQuestion
No.

START WRITING HERE
(Begin answer for each question on a new page)

Ans 1

$$(a) \quad f(w) = \frac{1}{2} \|Xw - y\|^2$$

$$o(w) = \bar{w} \bar{x}$$

$$E(w) = \frac{1}{n} \sum (t_d - o_d)^2$$

$$E = \left(\frac{\partial}{\partial w_0} + \frac{\partial E}{\partial w_1} \dots + \frac{\partial E}{\partial w_n} \right)$$

$$w_i = w_u + \Delta w$$

$$\Delta w = \eta \frac{\partial E}{\partial w}$$

↓
change in weight

$$\frac{\partial f}{\partial w} = \frac{1}{2}$$

Sr. No. 8526



START WRITING HERE

(Begin answer for each question on a new page)

Space
For
Marks

Question
No.

Ans 1

(b) overfitting occurs when model capacity is too high. This is because, more the capacity, there will be more variance. And high variance leads to overfitting.

Space
For
MarksQuestion
No.

START WRITING HERE
(Begin answer for each question on a new page)

Ans 2

a Input size = $H \times W \times C$
Filter size = $F \times F$

$$\text{Output} = \frac{\text{input size} - \text{filter size} + 2P}{S} + 1$$

$$= \frac{W - K + 2P}{S} + 1$$

$\left\{ \begin{array}{l} P = \text{padding} \\ S = \text{stride} \end{array} \right\}$

→ Because in modern architectures there is more depth, more complexity, more features, and more parameters and ~~that's~~ that's why depth wise separable convolutions is critical to

Space
For
MarksQuestion
No.

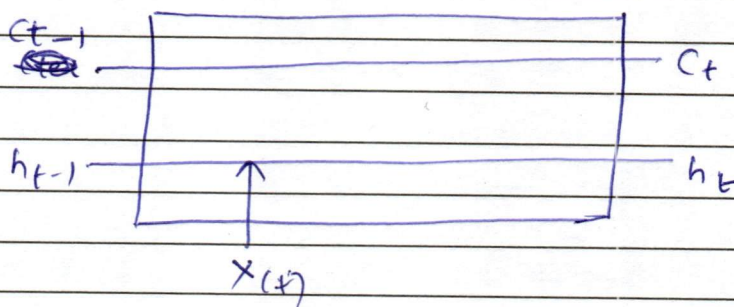
START WRITING HERE
(Begin answer for each question on a new page)

Ans 3

- (a) LSTM modifies this recursion, with the help of long term and short term memory. It has a long term memory to solve the problem of vanishing gradient. Also LSTM has 3 gates:
- 1) Forget gate
 - 2) Input gate
 - 3) Output gate.

On receiving an input in normal RNN, with time it would forget the initial timestamp information. But LSTM it has a long term memory. It has 2 states - a cell state and a hidden state. At the input received, the input gate decides what to store in the cell state and forget gate decides what part of the info is not at all required or relevant and can be forgotten. Then the relevant output at that particular timestamp is shown in the hidden state, and then the output is shown.

$$f(w_t[h_{t-1}, u_t] + b_t)$$



2 states; cell state and hidden state.

→ So previously, relevant info used to vanish on reaching further timestamp, losing context. But forget gate plays a crucial role, it decides what to forget and what not to based on multiple factors. Hence, saving the relevant info and preventing vanishing gradient.



Space For Marks	Question No.	START WRITING HERE (Begin answer for each question on a new page)
	Ans 3	
	(b)	GRU has 2 gates and 1 state. 1) Update gate 2) Reset Gate
		And it has 1 hidden state.
		The update gates role is to decide what input is required, and is relevant, as well what part of the memory is to be changed and showing the relevant output. whereas reset gate decides what to forget, and change the memory.
		So in the previous, LSTM, hidden state and candidate state would do the similar thing, take in the relevant info at that particular timestamp, store in the memory, and then the hidden state would display a part of the long term memory relevant at that particular timestamp.
		→ So this way GRUs update rule is combination of before hidden state and candidate hidden state.
		GRU have fewer parameters than LSTM, because it has less no. of states and gates than LSTM, it is lighter than LSTM.

Sr. No. 8526



8

Space
For
Marks

Question
No.

START WRITING HERE
(Begin answer for each question on a new page)

[illegible]

Sr. No.

8526



10

[illegible]

[illegible]

Sr. No. 38526



12

Space
For
Marks

Question
No.

START WRITING HERE

(Begin answer for each question on a new page)

[illegible]

Sr. No. 8526



14

Space
For
Marks

Question
No.

START WRITING HERE

(Begin answer for each question on a new page)

[illegible]

Sr. No. 8526



16

Question
No.

START WRITING HERE
(Begin answer for each question on a new page)